

Selection Optimism Is Tested Against the Wrong Null: A Corrected Procedure, and What Layer Selection Actually Costs

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Abstract

Audits of data leakage in machine-learning evaluation report what a selection step cost using the *selection-specific component*: the optimism gap at the selected candidate minus the mean gap across candidates. They then declare a selection effect when a bootstrap interval for that statistic excludes zero. We show the statistic is fine and the test is not. Writing $\Delta_{\text{sel}} = A - B$, with A the winner’s curse on the selection criterion and B the selected candidate’s advantage on data that played no part in selecting it, the null value of Δ_{sel} is A , not zero, because an argmax over a noisy criterion makes $\mathbb{E}[A] > 0$ whatever the data. In simulation with known ground truth the practised test has a type I error of 1.000 at every candidate count we tried: under a null in which selection buys nothing transferable, it declares an effect in all 2,000 audits. Worse, $\mathbb{E}[\Delta_{\text{sel}}]$ *falls* as real transferable quality rises, so the statistic is anti-monotone in the effect it is read as measuring. The correction is free, because audits already compute A : report the gap at the selected candidate, report A , and test B , whose null value is genuinely zero. That test is nominal (0.045–0.054 at $\alpha = 0.05$) and powered. We also ask what the existing instruments give here: against known ground truth, both Δ_{sel} and the direct held-out difference are unbiased for the optimism incurred, while Tweedie shrinkage and an argmax-shift correction — which must model how candidate quality is distributed — are biased and never reach nominal coverage in any configuration we tried (0.03–0.88 against 0.95). The estimator was never the problem; only the null was. Applied to hidden-state hallucination-detection probes selecting one of 32 layers, selection transfers real quality on AUROC and Brier and is undetermined on reliability and ECE, while the cost a practitioner pays — a threshold moved from reused to held-out data inflating achieved selective-prediction risk by 10–16% of target — is a reuse cost that needs no correction at all. We supply a five-mechanism leakage taxonomy, three audits verified against pinned third-party commits, and two results against our own interest.

1 Introduction

The protocol we study is hidden-state probing for hallucination detection: a linear probe is cheap to train, and reported AUROCs are often striking – 0.90 or higher, sometimes a perfect 1.000 – in a regime of few samples and many dimensions, where such numbers should invite scrutiny of the procedure that produced them.

Work that has scrutinized this protocol – adaptive data analysis, post-selection inference, and this paper’s own prior audit – measures its effect on *discrimination*, and summarises what the selection step cost with one statistic: the optimism gap at the selected candidate minus the mean gap across candidates. We take that statistic as the object of study, because it does not mean what it is read to mean. The selected index maximises the same criterion the gap contains, so the statistic is positive in expectation *even when no candidate carries transferable information*; an interval excluding zero therefore rejects a hypothesis nobody holds.

The reference it needs is already inside the audit. Writing $\Delta_{\text{sel}} = A - B$ — an identity, not a model — with A the winner’s curse on the selection criterion and B the selected candidate’s advantage on data that played no part in selecting it, the null value of Δ_{sel} is A . Since A is observable, the corrected statistic is $\Delta_{\text{sel}} - A = -B$, and the corrected test is a one-sample test of B , whose null value is genuinely zero. No permutation, no conditioning event, no added assumption.

How much this matters is an empirical question about the estimator, and we answer it with known ground truth. In simulation the practised test has a type I error of 1.000: under a null in which selection buys nothing transferable it declares an effect in all 2,000 audits, at every candidate count we tried, and the problem grows with the candidate set. The corrected test is nominal and powered. Worse for the practised statistic, $\mathbb{E}[\Delta_{\text{sel}}]$ *falls* as real transferable quality rises, because that quality cancels part of the curse — so it is anti-monotone in the effect it is read as measuring (§4.4).

Applied to our own harness the corrected procedure says selection recovers genuinely better layers on AUROC and Brier and leaves reliability and ECE undetermined. That changes the attribution but not the cost. The reused-versus-held-out gap is directly measured and needs no correction: a threshold calibrated on the former and transferred to the latter inflates achieved selective-prediction risk by 10–16% of target. What a practitioner pays is a *reuse* cost, and **a clean discrimination metric still does not imply a clean confidence estimate**. Our second claim is that measuring any of this is harder than it looks: a single case-study cell spans 1.44× across two admissible controls (§2) — severity is a property of a (mechanism, metric, *control*) *triple*.

A nested-cross-validation leakage bug in a probing pipeline we had full commit-history access to provided the seed case for this taxonomy; the question it raised – isolated or structural – defines the paper’s scope. We identify five *distinct* mechanisms by which evaluation labels leak into a reported metric, verify three against pinned commits of externally published pipelines, and measure each against a matched control (§5); most are small on the AUROC scale, one is not. A self-audit turns the same scrutiny on our own work: Mechanism 3’s primary synthetic estimate does not survive correction for its own construction asymmetries.

Contributions.

- **The test used to attribute optimism to selection has a type I error of 1.** The statistic’s null value is A , not zero, by the identity $\Delta_{\text{sel}} = A - B$. In simulation with known ground truth the practised test declares a selection effect in all 2,000 null audits at every candidate count, and the problem grows with the candidate set (§4.4).
- **Δ_{sel} is anti-monotone in the effect it is read as measuring:** its expectation falls as real transferable quality rises, because that quality cancels part of the curse.
- **A correction that costs nothing,** because audits already compute A : report the gap at the selected candidate, report A , and test B . The corrected test is nominal and powered (§4), and needs no conditioning event, selection region or added assumption.
- **What the imported corrections give here, measured.** Against known ground truth, Δ_{sel} and the direct held-out difference are both unbiased for the optimism incurred with near-nominal coverage, while Tweedie shrinkage and an argmax-shift correction are biased and never reach nominal coverage in any configuration (0.03–0.88 against 0.95). Winner’s-curse correction is the right tool when no held-out replicate exists; an audit always has one (§4.5).
- Applied to a 32-layer probe selection: selection transfers real quality on AUROC ($t = 9.07$) and Brier ($t = -5.69$) and is undetermined on reliability and ECE, while the practitioner’s cost is a *reuse* cost priced at 10–16% of a selective-prediction risk budget (§6.1).
- A five-mechanism taxonomy of label-information leakage in hidden-state probing pipelines, three verified against pinned third-party commits (§5), and a practitioner checklist (§7).
- **Results against our own interest:** a primary synthetic estimate that fails its own factorial control, two estimators we prove non-negative by construction and then decline to bootstrap, a second harness where the same procedure leaves calibration untouched, and a permutation null we built, showed redundant, and discarded (Appendix M).

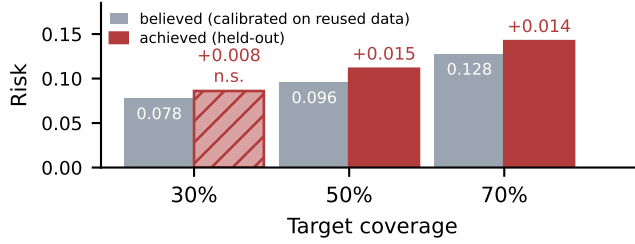


Figure 1: The consequence, before any mechanism-level detail. Grey is the selective-prediction risk a practitioner believes they bought by calibrating on reused data; red is what is incurred on held-out data at the identical threshold – 10–16% of the target risk, or 1,082 unflagged errors a day per 100,000 queries at a 70% coverage target, each arm priced at the coverage it actually achieves. The 30% cell is hatched: its BCa interval on the risk difference covers zero. **The selected object is a single integer**, which of 32 layers to report (Mechanism 2, §6.1).

What each claim is, and how strongly. Every claim sits in one tier of Table 1, read at that tier and nowhere at a stronger one; Table 6 in Appendix A assigns every individual result to one.

A Documented	Present in the harness as written; descriptive, no effect size claimed. Five mechanisms, three traced to pinned third-party source.
B Established	Interval excludes zero against <i>the right null</i> . The reused-versus-held-out gaps and the operating-point cost; <i>B</i> , the identified component of the decomposition, on AUROC and Brier; M1/M3/M5 severities.
C Not established	Reported, not relied on. <i>Every</i> selection-specific increment, which is tested against the wrong null by construction (§4); our primary synthetic estimate, which fails its own control; a non-reproducible prior number; a second harness showing no calibration effect.

Table 1: How to read every claim here. No prevalence estimate is given: repositories were found by targeted search, not sampled.

2 Related Work

Calibration, selective prediction, and conformal prediction. Our calibration-bridge finding (§6) connects selection bias to the calibration literature (Guo et al., 2017) and, via §6.1, to selective prediction (Geifman & El-Yaniv, 2017), which formalizes the risk-coverage tradeoff we use. Conformal prediction (Angelopoulos & Bates, 2023) and risk-controlling prediction sets (Bates et al., 2021) formally treat the question §6.1 asks informally – does a threshold calibrated on one sample control risk on another. We illustrate that question on selection-biased scores rather than proving a new risk-control result: those raw scores are themselves corrupted by the selection step audited here, a risk any conformal wrapper built on them inherits unchecked. To our knowledge none of these literatures has been connected to the leakage mechanisms in §5.

The definition we use. *Leakage is a change in the expectation of a reported metric attributable to evaluation-target information entering the estimation process.* The change is *control-relative*: severity is not a (mechanism, metric) *pair* but a (mechanism, metric, control) *triple* – the difference between the leaky procedure and *some* counterfactual, and defensible counterfactuals disagree. On one Mechanism-3 cell the two defensible controls – a blind retrain on the full fold, and the same retrain on the 85% subset its epoch count was chosen on – give +0.0034 and +0.0049, a $1.44\times$ spread; a contaminated adaptive control we disqualify on independent grounds gives +0.0198, making it $5.8\times$.

Adaptive data analysis and post-selection inference. Two of our five mechanisms – select a layer or threshold on the test set, then report at that choice – are the canonical setting of adaptive data analysis (Dwork et al., 2015). Our candidate-count relationship (§5) is shape-consistent with that bound, whose

worst-case guarantees exceed every severity we measure by two to three orders of magnitude. Cawley & Talbot (2010) show model-selection over-fitting induces bias growing with candidate count – Mechanism 2’s failure mode; new here is that the selected object is a layer index of a frozen network, contaminating a localization claim, not only a number. Both mechanisms have canonical priors outside ML (Ambroise & McLachlan, 2002; Varma & Simon, 2006, with nested CV as the remedy), and we claim no novelty on either discrimination result. What is new here is that the same two selection steps corrupt *calibration* and degrade the realized risk of a selective-prediction rule at a transferred threshold – an estimand neither line of work measures.

Existing leakage taxonomies and contamination. The closest prior art is Kapoor & Narayanan (2023, an 8-type taxonomy across 17 fields) and REFORMS (Kapoor et al., 2024, a 32-item checklist); Mechanism 1 is a direct case of their L1.3, feature selection under test-set visibility; Mechanism 4 applies the same structure to a layer index. The remaining three select against a validation signal rather than across the train/test boundary, and map onto neither taxonomy, since both treat train/test separation as the leakage surface, not iterative model selection against a validation signal. Roth (2026) quantifies four leakage classes across 2,047 datasets, but in discrimination only. Hussain & Kantarcioglu (2026, PARALLAX) find four of six corpora leak the ground-truth answer directly into the prompt, via a mechanism distinct from all five here; pretraining-data contamination (Shi et al., 2024) is different still: what the model has already seen, not whether a selection step used test-set signal afterward.

3 Method

Two case studies come from our own prior work, with full code, data, and commit-history access. The external ones required a real, runnable public repository verified via the host, a reported AUROC ≥ 0.90 , and an evaluation protocol whose leakage risk we could assess from the code.

Evidentiary status. Three case studies are *fully verifiable here* – checkpoint selection (MultiHaluDet), best-layer selection (HallucinationPatternDetection), and test-set threshold selection (Mechanism 5, audited in *both* of the above repositories at the same pinned commits) – all external, published, code-available systems, with every script and result JSON released; the third is detailed with its calibration-contamination bug in §O. One (CV-based layer selection) is *verifiable from a derived artifact*: our own prior pipeline, whose raw hidden-state cache is too large to ship, though a small derived artifact lets a reader recompute the primary numbers. One (full-dataset probe reuse) is *not independently re-verifiable* – its number comes from our own prior work with no surviving scripts; we keep it because the mechanism is real and is this paper’s origin, but never compare it against the code-verified numbers.

Severity is not one scale. Four named quantities avoid conflation: Δ_{synth} (calibrated synthetic-harness gap at a fixed Bayes-optimal operating point), Δ_{real} (the same on real Mistral-7B/HaluEval features), Δ_{sel} (within-split selection-specific component against its own permutation null), and Δ_{boot} (bootstrap max-bias over already-published AUROC arrays, no model retrained) – all AUROC differences, sharing units but not a reference. BCa (bias-corrected and accelerated bootstrap) 95% intervals are the default decision rule (established when the interval excludes zero) except for one case study, whose two candidate estimators are provably non-negative by construction, exactly the condition under which BCa’s correction cannot restore coverage (simulation confirms 0% empirical coverage there), so we never use BCa as evidence for that case study.

What the intervals are about. Every interval in this paper resamples the *replicate* – one split or seed draw – not the example, and every replicate is a resplit of the same underlying sample. The estimand is therefore *split-conditional*: the expected optimism of this selection procedure on this harness, over the randomness of the analyst’s split. It is not a population quantity, and no interval here licenses extrapolation to another dataset, model or feature representation. That is a real limit, not a formality: §6 reports a harness where the same procedure produces no calibration effect at all. Read “established” throughout as “established for this harness” (Appendix B).

Multiple-comparison correction. Where we additionally compute Holm-Bonferroni across cells tested together (Table 8), it eliminates the marginal finding in most families we checked. Applying family-wise control, no marginal finding survives outside one mechanism’s non-marginal threshold-selection result (a single large effect outside any multi-cell family) and the table’s exceptions (two families, three cells). The robustness analyses added in Appendices I–L – the binning sweep, the placebo decomposition, the regularization sweep and the second-harness replication – are **not** family-wise corrected. They were run to stress one already-reported finding rather than to search for new ones, and we report every cell of each, including the ones that go against us; but they should be read as robustness checks on a single claim, not as an independent family of confirmations.

4 The Estimand, Its Null, and a Corrected Test

This section is the paper’s methodological core. It concerns not the audits that follow but the statistic audits use, and it applies wherever a candidate is chosen by argmax of a noisy criterion and the choice is then charged with the resulting optimism.

4.1 The statistic, and the test applied to it

Let $l = 1, \dots, K$ index candidates (layers, checkpoints, thresholds). Let cv_l be the criterion computed on the fold used for selection and ho_l the same metric on genuinely held-out data, and let $\text{gap}_l = cv_l - ho_l$ and $l^* = \arg \max_l cv_l$. The quantity used to separate what *selection* cost from what ordinary reuse cost is the selection-specific component

$$\Delta_{\text{sel}} = \text{gap}_{l^*} - \overline{\text{gap}}. \quad (1)$$

The practice we are questioning is not the statistic but the test: an audit computes a bootstrap interval for Δ_{sel} over replicates and declares a selection effect when that interval excludes zero.

4.2 Its null value is A , not zero

Write

$$\Delta_{\text{sel}} = A - B, \quad A = cv_{l^*} - \overline{cv}, \quad B = ho_{l^*} - \overline{ho}. \quad (2)$$

This is an identity, not a model: substituting $\text{gap}_l = cv_l - ho_l$ into (1) and expanding gives (2) directly. Its consequence is the whole point. Under a null in which candidates carry no *transferable* quality, l^* is uninformative about ho , so $\mathbb{E}[B] = 0$; but l^* maximises cv , so $\mathbb{E}[A] > 0$ strictly whenever the criterion is noisy and $K > 1$. Therefore

$$\mathbb{E}_{H_0}[\Delta_{\text{sel}}] = \mathbb{E}[A] > 0.$$

The null value of Δ_{sel} is A . Testing it against zero tests a hypothesis that is false whenever an argmax is applied to anything noisy, and rejecting it is not evidence that selection did anything.

We stress what this is *not*. Δ_{sel} is not biased for its own estimand, and nothing here requires a permutation, a conditional distribution, or a bootstrap of a re-parameterisation. A is computable from quantities the audit already has.

4.3 The corrected test

Because A is observable, the corrected statistic is $\Delta_{\text{sel}} - A = -B$. Equivalently, and more directly:

Test B . $B = ho_{l^*} - \overline{ho}$ is built entirely from held-out values, which played no part in choosing l^* . Its null value is genuinely zero, so a one-sample t -test of B over the audit’s own replicates is valid as it stands.

The procedure for an auditor is then three lines. Report gap_{l^*} , the optimism actually paid at the reported configuration, which is directly measured and needs no correction. Report A , the winner’s curse on the selection criterion, which is the reference Δ_{sel} must beat. Test B , not Δ_{sel} , if the question is whether the selected candidate carries transferable quality.

4.4 Size and power

Whether this matters is an empirical question about the estimator, so we answer it with known ground truth rather than with our own data. For K candidates we draw true qualities $q_l \sim N(0, \tau^2)$, observe $cv_l = q_l + N(0, \sigma^2)$ and $ho_l = q_l + N(0, \sigma^2)$ with $\sigma = 0.03$, form l^* , and run 2,000 independent audits of 50 replicates each. At $\tau = 0$ no candidate carries transferable quality, so the rejection rate there *is* the type I error. Table 2 reports both tests.

K	τ	Declares a selection effect		Expected values		
		as practised	corrected	Δ_{sel}	A	B
8	0.000	1.000	0.045	0.0427	0.0427	-0.0000
8	0.010	1.000	0.197	0.0407	0.0450	0.0043
8	0.040	0.934	0.993	0.0260	0.0710	0.0450
32	0.000	1.000	0.054	0.0620	0.0620	0.0001
32	0.010	1.000	0.310	0.0589	0.0655	0.0066
32	0.040	0.990	1.000	0.0376	0.1031	0.0656
128	0.000	1.000	0.053	0.0778	0.0779	0.0001
128	0.010	1.000	0.444	0.0738	0.0820	0.0083
128	0.040	0.995	1.000	0.0465	0.1299	0.0835

Table 2: Rejection rates at $\alpha = 0.05$, 2,000 audits per row. At $\tau = 0$ the rate is the type I error. The practised test rejects on *every* null audit at every K ; the corrected test is nominal (0.045–0.054) and retains power as τ grows. `111_estimator_size_power.py`.

Three things follow.

The practised test has a type I error of 1.000. Not inflated — saturated, at every candidate count we tried. Under a null in which selection buys nothing transferable, an audit following current practice declares a selection effect in every single one of 2,000 simulated studies. This is not a small-sample artifact: the rejection is driven by $\mathbb{E}[A] > 0$, which grows with K (0.043 at $K = 8$, 0.078 at $K = 128$), so the problem worsens exactly as the candidate set grows and the audit becomes more worthwhile.

The corrected test is nominal and powered. Type I error 0.045–0.054 against a nominal 0.05, with power rising to 1.000 by $\tau = 0.04$ at $K \geq 32$.

The practised statistic moves the wrong way. As real transferable quality increases, $\mathbb{E}[\Delta_{\text{sel}}]$ *falls* (0.062 to 0.038 at $K = 32$) while $\mathbb{E}[B]$ rises. Δ_{sel} is anti-monotone in the quantity an auditor is trying to detect, because real transferable quality cancels part of the curse. An audit that reads a *larger* Δ_{sel} as a worse selection problem has the sign of the relationship backwards.

4.5 What the existing corrections give here, and what they cost

The referee’s question for any such critique is the right one: statistics has instruments for selection bias already, so what do they give in this setting? We answer empirically rather than by assertion. Using the same simulation, we estimate the quantity an auditor actually wants — the optimism incurred at the reported configuration, $\theta = cv_{l^*} - q_{l^*}$, knowable here because q is — with four estimators, and report bias, dispersion, RMSE and interval coverage (Table 3).

Two things follow, and the first is a correction to our own framing.

The estimator was never the problem. Both Δ_{sel} and the direct gap are essentially unbiased for θ (bias $\leq 10^{-4}$) with near-nominal coverage at every τ . Nothing is wrong with reporting Δ_{sel} as a point estimate of the optimism incurred. What is wrong is testing it against zero, because $\theta > 0$ whenever an argmax acts on a noisy criterion — which is why that test has type I error 1 in §4.4. The test asks whether optimism exists; the answer is always yes; the informative question is whether the selected candidate *transfers*, and that is B .

τ	Estimator	Bias	SD	RMSE	Coverage	Uses held-out?
0.00	Δ_{sel} (as reported)	−0.00006	0.0045	0.0042	0.966	no
	direct gap $\text{cv}_{l^*} - \text{ho}_{l^*}$	−0.00003	0.0047	0.0042	0.974	yes
	Tweedie / empirical Bayes	−0.00547	0.0016	0.0057	0.046	no
	argmax-shift correction	−0.00007	—	0.0021	n/a	no
0.01	Δ_{sel} (as reported)	+0.00011	0.0048	0.0042	0.974	no
	direct gap	+0.00011	0.0049	0.0042	0.977	yes
	Tweedie / empirical Bayes	−0.00254	0.0016	0.0034	0.588	no
	argmax-shift correction	+0.00315	—	0.0041	n/a	no
0.03	Δ_{sel} (as reported)	+0.00003	0.0080	0.0043	0.980	no
	direct gap	+0.00003	0.0080	0.0043	0.983	yes
	Tweedie / empirical Bayes	+0.00143	0.0038	0.0082	0.291	no
	argmax-shift correction	+0.01816	—	0.0194	n/a	no

Table 3: Estimating θ at $K = 32$, 4,000 audits of 50 replicates, nominal coverage 0.95. The argmax-shift correction is deterministic within an audit, so it has no within-audit interval. `113_estimator_comparison.py`.

Winner’s-curse correction buys nothing here, and costs coverage. Both cv-only corrections must model how candidate quality is distributed, and they pay for it. The Tweedie estimator has the smallest SD of any estimator in the table and the worst coverage by a wide margin — 0.046 against a nominal 0.95 at $\tau = 0$, 0.588 at $\tau = 0.01$, and never above 0.883 in any of the nine (K, τ) cells we ran — because shrinkage buys precision with a bias its interval does not acknowledge. The argmax-shift correction is unbiased only when candidates are genuinely exchangeable and drifts to +0.018 once they are not. Neither is competitive with simply differencing against held-out data.

This is why we do not reach for conditional selective inference or a winner’s-curse correction. Those instruments exist because in their home settings — a genome-wide scan, a lasso fit — there is no held-out replicate of the selected effect. An audit always has one, which is what makes the direct difference unbiased with no model for q . Where held-out data is unavailable they are the right tool and stronger than anything here; where it is available they are a worse answer to an easier question.

4.6 Positioning

That selection on a noisy criterion biases the selected estimate is old. Valid post-selection inference (Berk et al., 2013) gives simultaneous intervals over selection outcomes; conditional selective inference (Fithian et al., 2014; Lee et al., 2016) conditions on the selection event for exact intervals; false coverage-statement rate control (Benjamini & Yekutieli, 2005) adjusts intervals for selected parameters; and a genetics literature corrects winner’s-curse point estimates directly (Zöllner & Pritchard, 2007; Sun & Bull, 2005; Zhong & Prentice, 2008), with Efron (2011) giving the empirical-Bayes view. Bates et al. (2024) is nearest in spirit, asking what cross-validation estimates and showing naive CV intervals undercover.

We claim no improvement on any of them. Our contribution is narrower and, we think, more useful to the practitioners concerned: the audits in question already compute A without recognising it, so the fix is free, and §4.5 shows the imported alternatives are worse in this setting rather than better. The absence of that observation is what allows a test with type I error 1 to remain standard.

5 Five Mechanisms, Measured

Three axes locate the five mechanisms: (a) the provenance of the statistic selected on — full-dataset labels, an inner CV fold, or the test set; (b) the object selected — a feature, a layer, a checkpoint, or a threshold; and (c) whether the selection’s output is reused downstream as a feature. These are documented instances, not an orthogonal partition.

Table 7 (Appendix C) gives every mechanism’s severity traced to source code, one row per mechanism (two each for Mechanisms 1 and 3, whose harnesses disagree or differ in provenance). Mechanism 1’s narrative number (+0.19, from prior work) is excluded from every cross-mechanism comparison and shown only for orientation; its *second* row is a new controlled synthetic reconstruction that *is* code-verified and is 5.1–11.4× above every Mechanism 2–4 estimate. Every cross-mechanism severity comparison in this paper pools only Mechanisms 2–4.

Two numbers are as much about measurement as about leakage. Mechanism 3’s *primary* synthetic estimate does not survive its own control: a $2 \times 2 \times 2$ factorial crossing selection-set size, selection-run budget, and training depth carries the as-published control’s +0.0030 to a fully corrected −0.0001 (BCa [−0.0021, +0.0021], $p = 0.657$), attributing 75.5% of the movement to selection-set size at the shipped early-stopping fraction – a dominant factor, not a fixed one (Appendix B). The decomposition, not the null, is the contribution: it names the construction choice that manufactures a false positive of this size. Mechanism 4’s estimate rests on a statistic non-negative by construction (Appendix D states and proves both), so its sign carries no information; against a permutation null, no cell is significant above it. Mechanism 5’s F1 severity is n -dependent ($\approx 6.3\times$ smaller at $n=2000$ vs. $n=140$); whether it stays distinguishable from noise at typical benchmark sizes is untested here.

What governs severity within a harness. Severity rises monotonically and concavely with candidate count at a fixed operating point and vanishes near ceiling, but neither relationship transports: an exchangeable-candidate replication at adequate power does not reproduce the first, and the second is exceeded by an order of magnitude at a matched operating point in our own real-feature harnesses. A matched comparison shows something beyond these two covariates is unabsorbed, but cannot separate mechanism identity from the harness maturity correlated with it, so the only conclusion it licenses is negative. Numbers and fits in Appendix N.

6 Operational Consequences: What the Reuse Gap Costs a Deployment

A selective-prediction threshold calibrated on reused-fold scores buys a risk budget the deployed system does not meet (Table 5): choosing one of 32 layers by argmax inflates it by 10–16% relative. This is the paper’s most robust finding, because it is a directly measured transfer cost rather than a decomposed attribution: at the 50%-coverage point it survives regularization, as does the reuse gap that drives it, unlike Mechanism 1’s ECE gap (Appendix K). We first show why, then that it is not limited to one construction.

Every severity number so far is a gap in AUROC or F1 – discrimination metrics, which ask only whether a probe ranks positives above negatives correctly. None of them ask whether the probe’s *probability outputs* are honestly reliable: whether a prediction made at 0.9 confidence is right about 90% of the time. This is a different question, and the literature on selection bias in evaluation (adaptive data analysis, post-selection inference, and this paper’s own §5) has not asked it for this failure class. We do, on two structurally distinct mechanisms already audited above, using their own already-computed scores – no new data, no new model calls, only a new metric applied to existing LEAKY/CLEAN_MATCHED comparisons (defined below).

Method. For Mechanism 2 (CV-argmax layer selection), we take the CV-selected layer’s own probability outputs on the reused selection fold (LEAKY) against the same layer’s probability outputs on the genuinely held-out split (CLEAN_MATCHED), across the same 50 randomized-split repetitions §5 already uses, replicating that section’s layer-selection rule exactly (verified: selected layers match to the rep). For Mechanism 1’s real-feature analog, we take the same full-dataset-fit probe’s in-sample probability outputs (LEAKY) against a 5-fold out-of-fold refit (CLEAN_MATCHED), on the same cached real Mistral-7B/HaluEval features and probe convention as its AUROC comparison (replication verified to four decimal places against that AUROC number). For both, we report Brier score (primary; well-defined without binning at these sample sizes) and 10-bin expected calibration error (ECE; secondary, standard in the calibration literature), with the same BCa 95% bootstrap interval this paper uses throughout.

Mechanism	Brier gap (BCa 95%)	ECE gap (BCa 95%)
2 (CV-argmax layer selection)	+0.0082 [+0.0030, +0.0137]	+0.0171 [+0.0099, +0.0242]
1 (full-dataset probe reuse, real features)	+0.0260 [+0.0252, +0.0269]	+0.0193 [+0.0184, +0.0203]

Table 4: Calibration gap (CLEAN_MATCHED minus LEAKY; positive means the leaky condition looks more reliable than it honestly is). Mechanism 2: 50 randomized splits, from a cached pipeline whose raw hidden-state cache is too large to ship, though a derived artifact lets a reader recompute these numbers. Mechanism 1: $n=400$, computed on the cached real-feature analog – *not* the non-reproducible prior-work number of §3, which is excluded here as everywhere. All four cells survive Holm-Bonferroni jointly with Table 5.

Result. Both gaps are positive and both intervals exclude zero for both mechanisms (Table 4). Mechanism 2 carries the claim: an argmax over 32 candidate layers, with no fitting on the evaluation data at all, still moves Brier by +0.0082 and ECE by +0.0171, both BCa-established. **Both replicate on a disjoint HaluEval pool of the same size** (+0.0112 and +0.0200, each established and larger); both pools are Mistral-7B, so the *calibration* gap is established across question sources on one model, and the selection step itself across architectures next.

Brier is not a calibration metric; its reliability term is. A Brier gap mixes calibration with discrimination, and argmax-AUROC selection optimises discrimination directly, so that gap cannot on its own carry a calibration claim. We decompose it on the same 50 splits, $BS = REL - RES + UNC$. The **reliability** term carries the effect (+0.0046, BCa [+0.0027, +0.0065]); **resolution** does not (+0.0028, BCa [−0.0014, +0.0070], covering zero); the uncertainty term moves +0.0009 and reflects a matched base rate (0.765 reused, 0.763 held out) rather than an effect. Two binning-free statistics agree: the calibration slope falls from 0.6807 to 0.6185 (gap +0.0621, BCa [+0.0145, +0.1033] – both arms overconfident, the held-out arm more so), and a binning-free *isotonic calibration error* – $\mathbb{E}[(p - \hat{g}(p))^2]$ with $\hat{g}$ an isotonic recalibration map (Zadrozny & Elkan, 2002), a plug-in estimate of the squared L_2 calibration error in the family surveyed by Blasiok et al. (2023) – gives +0.0034 (BCa [+0.0020, +0.0050]). We therefore describe the Brier result as a degradation in probabilistic prediction quality and rest the calibration claim on the reliability term.

The ECE gap is not an artifact of binning. Binned ECE is a biased and binning-sensitive estimator of calibration error at these sample sizes (Vaicenavicius et al., 2019; Kumar et al., 2019; Roelofs et al., 2022), and distribution-free guarantees for binned calibration require care about the binning itself (Gupta & Ramdas, 2021), so a single setting cannot carry the result. Sweeping bin count $\in \{5, 10, 15, 20\}$ across equal-width and equal-mass schemes, all eight settings give a positive gap whose interval excludes zero, spanning +0.0164 to +0.0212. The submitted 10-bin equal-width value, +0.0171, sits mid-range rather than at a favorable extreme. The binning-free smooth calibration error reported above agrees in sign and significance, which is the check that does not inherit the estimator’s bias.

How much of this is selection? Neither the placebo nor the null supports attributing it. Two references are needed, and the paper’s claim fails against both. First, a placebo: the comparison above evaluates one layer on two samples, so it could reflect an ordinary reused-versus-held-out difference that any layer would show. Repeating the identical 50 splits at every one of the 32 layers gives a placebo that is real but smaller (reliability +0.0020, BCa [+0.0006, +0.0034]), leaving a *selection-specific increment* of +0.0026 (BCa [+0.0014, +0.0040]) for reliability and +0.0099 (BCa [+0.0054, +0.0150]) for ECE. Second – and decisively – §4 says those increments must not be tested against zero at all. Applying the corrected procedure instead (Table 16), B is not established on reliability ($t = -1.33$) or on ECE ($t = -0.18$), and is established on Brier (-0.0066 , $t = -5.69$) in the direction that favours transfer. **We therefore claim no selection-specific calibration effect**, and note that testing Δ_{sel} against zero would have declared one on every metric. What the decomposition does identify is favourable to selection: the component measured only on data that played no part in choosing the layer points the same way on all three losses, so the selected

layer is if anything *better* calibrated on fresh data, established for Brier. The reused-versus-held-out gap itself is untouched by any of this; it is directly measured, and §6.1 prices it. It is a cost of *evaluating on reused data at a selected configuration*, and that is how we report it.

Across architectures. Re-running the argmax protocol on cached states from four other architectures – identical n , split, folds, reps, seed and probe – the selection-specific component is positive on all four and excludes zero on three (Table 12); those caches are forced-choice rather than generation-labeled and test discrimination only. We note this for completeness and lean on it for nothing: by §4 a positive selection-specific component is what this estimator returns under its own null, and we did not compute per-architecture nulls.

Regularization is not a competing explanation for the gap. Mechanism 1’s probe interpolates at $C=1.0$, and **its ECE gap does not survive $C=0.01$** – a fact we report rather than set aside. Mechanism 2 is different: it never fits on the reported data, and at a standardized probe that cannot interpolate the reuse gap is undiminished (reliability $+0.0056$ against $+0.0046$ at the shipped probe; ECE $+0.0206$ against $+0.0171$), with the risk violation unchanged at $+0.016$ (Appendix K). We make no claim here about the selection-specific *increment* under regularization: we computed its mechanical null only at the shipped probe, and without a null per probe configuration the increment is not interpretable (Appendix M).

Where it does not hold. We establish calibration on two of the five mechanisms and on one feature representation. Run on real features from two model families, this same construction inflates discrimination ($\Delta_{\text{sel}} = +0.0070$ and $+0.0072$, established in both) *without* corrupting calibration: the selection-specific calibration increment is null or significantly negative, and the risk violation is not established (Appendix L). **The calibration consequence does not follow automatically from the discrimination one.** Where it does hold, the selection step that inflates AUROC also makes the resulting confidence estimates look more trustworthy than they are.

6.1 Does this move a deployment’s actual error budget?

A calibration gap alone leaves open whether it has operational stakes. We use the selective-prediction framework (fix a confidence threshold, abstain below it, and rely on the error rate among non-abstained predictions) to check directly: if a practitioner calibrates a deployment threshold using the same reused-fold (LEAKY) scores this paper already shows are optimistic, what risk do they actually get when that identical threshold is applied to genuinely new (CLEAN_MATCHED) data?

Method. At each of three target coverage levels (30%, 50%, 70%) we take the threshold achieving that coverage on LEAKY and LEAKY’s own reported risk there – what a practitioner using only the reused fold would believe – then apply the identical numeric threshold to CLEAN_MATCHED. Risk violation is actual minus believed, BCa 95% as elsewhere.

Result. Figure 1 plots the risk–coverage curves behind Table 5. **Mechanism 2 carries this result**, and it is the harder case: the only quantity chosen on the reported data is a single integer, which of 32 layers to use. Its violation is established at 50% and 70% coverage and not at 30%, where the BCa interval covers zero before any correction – a 10–16% relative inflation of the target risk.

Pricing the transfer at the coverage actually achieved. A threshold chosen for a 70% coverage *target* on the reused fold does not deliver 70% coverage once the score distribution shifts; it delivers 70.71%. Multiplying a risk difference by a fixed answered-query count therefore misprices the transfer, so we price each arm at the coverage it achieves. At a 70% target a practitioner has bought 0.128 risk over the 70,000 queries they expect to answer, or 8,945 errors; the system answers 70,707 at 0.142, or 10,027 – 1,082 **additional unflagged errors daily** (BCa [435, 1718]), against 972 under the fixed-coverage estimator, and 825 at 50%. Coverage shifts furthest at the 30% target, where the coverage-aware excess of 368 errors excludes zero, [77, 662], and the fixed-coverage estimator’s does not; we flag that cell as established *only* under the coverage-aware count and keep the risk-difference verdict as the conservative one (Appendix I). What

Mech.	Cover- age	Believed risk	Actual risk	Violation (BCa 95%)
2	30%	0.078	0.086	+0.008 [−0.001, +0.018] (<i>n.s.</i>)
2	50%	0.096	0.110	+0.015 [+0.007, +0.023]
2	70%	0.128	0.142	+0.014 [+0.005, +0.023]
1	30%	0.000	0.002	+0.002 [+0.001, +0.004]
1	50%	0.000	0.007	+0.007 [+0.006, +0.008]
1	70%	0.000	0.009	+0.009 [+0.008, +0.010]

Table 5: Selective-prediction risk violation: risk a practitioner believes they have (from LEAKY) versus risk actually achieved (on CLEAN_MATCHED) at the identical threshold (n as in Table 4). The violation column is the paired per-split mean, so it need not equal the difference of the two rounded risk columns. These six rows and Table 4’s four form one 10-cell family; nine are established per-cell and all nine survive Holm-Bonferroni correction. **Three of the nine rest on a construction we disqualify elsewhere:** Mechanism 1’s believed risk is exactly zero, so its violation *is* the achieved risk – non-negative by construction, where §3 declines BCa as evidence. **Six carry no such caveat.**

degrades is the abstention rate a deployment was sized against; this transfers an empirically chosen threshold and neither constructs nor contradicts a distribution-free risk-controlling procedure. The per-day counts scale one dataset’s split-to-split behaviour to a hypothetical query volume: they convey magnitude, not a population estimate.

Mechanism 1, and robustness. With a candidate set of size one, selection reduces to fitting on the evaluation data: believed risk is *exactly zero* at every coverage against an actual 0.2–0.9%. That is the largest cell in the paper and the least informative – non-negative by construction, not independent of §6’s Brier gap, and an already-documented error – so we retain it as the degenerate case the other four are weaker versions of and base no claim on it. Repeating the analysis at $C = 0.01$, away from exact interpolation, leaves the risk violation intact (+0.0079, BCa [+0.0070, +0.0090] at 50%, against +0.0075 at $C = 1.0$ – if anything marginally larger). The consequence therefore does not depend on the near-interpolation regime that shrinks *Mechanism 1*’s raw Brier and ECE gaps. Mechanism 2 behaves differently again, because it never fits on the reported data at all: under the same regularization its selection-specific calibration increment grows rather than shrinks (Appendix K).

An externally verified instance. One of the audited pipelines computes its own ECE with a threshold-independent confidence but a threshold-dependent correctness, the threshold being its own test-label-selected Youden cut: the pipeline’s reported ECE is contaminated by the mechanism we audit it for. This is a code-level defect at a pinned commit rather than a decomposed attribution, so the null question of §4 does not bear on it. Detail, magnitude and its single- source caveat in Appendix O.

7 A Checklist for This Literature

Five questions cover the five mechanisms above, in checklist rather than taxonomy order (full version with per-item fixes in supplementary material):

1. (M1) Is any feature computed by fitting on the full dataset’s labels before an outer CV loop treats part of that dataset as held out?
2. (M3) Inside a CV fold, is a “best” checkpoint/epoch chosen on the same fold whose predictions are reused downstream as clean out-of-fold output?
3. (M4) Is a “best” layer/config selected by argmax of a metric computed on the same test set later reported as the headline number, across many candidates, with no correction?
4. (M2) Was a hyperparameter chosen by maximizing a cross-validated metric, with that same CV number reported as performance and no genuinely held-out check?

5. (M5) Was a decision threshold chosen by optimizing a metric on the test labels themselves, with that same test set used to report the metric at that threshold?

Q1–Q5 name the mechanism and where to look; three are pinned-commit-verified. Q5 catches what nothing upstream will: AUROC ignores threshold selection, so an AUROC-only audit misses inflated F1/accuracy. The checklist supplies no severity *number* – §5’s surface is not a calculator (matched measurements exceed its prediction by 9.2–24.3×). Measure severity in your own harness against a matched control.

Is the checklist automatable by static analysis? Not at the line level, and the failure mode is diagnostic. A regex scanner for four of the five patterns, run against 7 repositories (2 audited, 5 externally found), produced 7 raw hits, all false positives on manual reading, and missed both known true positives – a keyword-list gap, and an argmax-plus-test-variable design that cannot fire when the test split lives in another module. A blinded language-model re-read agreed with all 7 classifications and located both missed bugs unprompted. Line-level regex matching is the wrong tool: selection statements relate to what they select on, often across files – not evidence the class resists static detection.

The prescribed fixes are cheap. Nesting cross-validation removes +0.0232 AUROC of Mechanism 2’s optimism (one run, within 1 SD of Table 7’s +0.0255) in seconds of CPU time; constructing Mechanism 3’s control fairly – matching the honest arm’s selection-set size to LEAKY’s, worth 75.5% of the +0.0030 → −0.0001 movement against 28.1% for the disjoint held-out set – costs under 5 minutes, with a +0.0006–+0.0030 residual surviving on real features. Both are measured against their naive alternatives, not merely prescribed.

8 Conclusion

The statistic this literature uses to say what a selection step cost is sound; the test applied to it is not. Its null value is the winner’s curse on the selection criterion, not zero, and that follows from an identity rather than a model. The consequence is not a mild miscalibration: in simulation the practised test declares a selection effect in every null audit at every candidate count we tried, and the statistic’s expectation moves *against* the effect it is read as measuring. The correction costs nothing, because the reference is already computed inside the audit. Report the gap at the selected candidate, report the curse, and test the transferred component.

Applied to a 32-layer probe selection, that procedure says selection recovers genuinely better layers on AUROC and Brier, leaves reliability and ECE undetermined, and reattributes the optimism a practitioner pays: it is the cost of evaluating on reused data at a selected configuration, which shows up as a 10–16% inflation of a selective-prediction risk budget and needs no correction to measure.

We held this paper to the standard it asks of the literature it audits. Our primary severity estimate does not survive its own factorial control; we prove two estimators non-negative by construction and then decline to bootstrap them; a second harness leaves calibration untouched; the selection attribution an earlier version of this work asserted is a claim we withdraw here; and the permutation null we first built to withdraw it turned out to be redundant, so we report that too. An audit of evaluation protocols that exempts itself measures nothing.

Limitations

The result about the test is general; the applied numbers are one harness. Every interval resamples splits of a single sample, so those estimands are split-conditional and nothing here extrapolates to another dataset, model or representation — and a second representation shows no calibration effect at all. The simulation of §4.4 uses Gaussian candidate noise with equal variance on both arms; it demonstrates that the practised test is invalid, not the exact size it would have under any particular real harness. Where exact conditional selective inference is available it is stronger than what we propose, and we do not compare against it empirically. Mechanism 1’s ECE gap does not survive regularization, the severity surface is a fit not a bound, one case study’s headline number is not independently re-verifiable, and we give no prevalence estimate: repositories were found by targeted search, not sampled. Appendices P–R have the rest.

AI Use Statement

In this work, we used generative AI tools (Anthropic Claude, accessed through a coding assistant) for several tasks carrying *required* disclosure: designing and providing feedback on research methodology and experiments, implementing methods, and interpreting results. Concretely, the assistant proposed and implemented the placebo control that decomposes the calibration gap into a selection-specific increment and an ordinary-reuse baseline (§6); the Murphy decomposition, calibration slope/intercept and isotonic calibration error used to separate reliability from resolution; the eight-setting ECE binning sweep; the coverage-aware deployment error count that replaced a fixed-coverage estimator (§6.1); the regularization analysis of Appendix K; and the second-model-family replication of Appendix L, whose result is negative and is reported as such. It also proposed and implemented the analysis that is this paper’s central result: the observation that the selection-specific component is tested against the wrong null, the exact $A - B$ decomposition, the identification of B as the estimable component, the size-and-power simulation of §4.4, and the corrected report of Appendix M. That analysis withdrew a claim an earlier version of this work asserted, and the paper was restructured around the negative result. It also built the permutation null we subsequently showed redundant, and identified the redundancy; we report both the construction and its retraction rather than quietly deleting the latter. It also drafted and edited substantial portions of the text, produced Figure 2, and proposed the current title. We did not use generative AI tools to generate the synthetic datasets or to formulate the non-negativity results of Appendix D, whose statements and proofs are the authors’ own. Support for qualitative and thematic data analysis and assistance with translation are not applicable to this work.

We have reviewed all AI-assisted work, and the review was mechanical rather than impressionistic. Every numerical claim in this paper is checked by an automated verifier that recomputes the value from the JSON emitted by the script that produced it and asserts its literal presence in the $\text{\LaTeX}$ source; the verifier currently runs 178 such checks and fails the build on any mismatch. It caught a transcription error in a reported figure during preparation of this version. The analysis scripts import the original study’s split generator, cross-validation object and probe-fitting routine rather than reimplementing them, so the protocol is inherited rather than re-derived, and one arm of Appendix K is replayed from the cached artifact specifically so that it reproduces an independently computed result as a consistency check. Where an AI-designed analysis returned a result unfavorable to our claim — the placebo, which attributes roughly 43% of the gap to ordinary reuse, and the second-model-family replication, which does not reproduce the calibration effect at all — that result is reported in the abstract and body rather than relegated or omitted.

We take responsibility for the final content of this work, including text, claims, or artifacts produced with the aid of generative AI.

References

- Alvi, R., Sayeedi, N. L., & Sayeedi, M. F. A. (2026). MultiHaluDet: Multilingual Hallucination Detection via LLM Hidden State Probing. In *Proceedings of the 1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM 2026)*.
- Ambroise, C., & McLachlan, G. J. (2002). Selection Bias in Gene Extraction on the Basis of Microarray Gene-Expression Data. *Proceedings of the National Academy of Sciences*, 99(10), 6562–6566.
- Angelopoulos, A. N., & Bates, S. (2023). Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4).
- Bates, S., Angelopoulos, A., Lei, L., Ryan, J., Malik, J., & Jordan, M. (2021). Distribution-Free, Risk-Controlling Prediction Sets. *Journal of the ACM*, 68(6).
- Blasiok, J., Gopalan, P., Hu, L., & Nakkiran, P. (2023). A Unifying Theory of Distance from Calibration. *Proceedings of the 55th Annual ACM Symposium on Theory of Computing (STOC 2023)*.
- Benjamini, Y., & Yekutieli, D. (2005). False Discovery Rate-Adjusted Multiple Confidence Intervals for Selected Parameters. *Journal of the American Statistical Association*, 100(469), 71–81.

- Berk, R., Brown, L., Buja, A., Zhang, K., & Zhao, L. (2013). Valid Post-Selection Inference. *The Annals of Statistics*, 41(2), 802–837.
- Brier, G. W. (1950). Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, 78(1), 1–3.
- Bates, S., Hastie, T., & Tibshirani, R. (2024). Cross-Validation: What Does It Estimate and How Well Does It Do It? *Journal of the American Statistical Association*, 119(546), 1434–1445.
- Cawley, G. C., & Talbot, N. L. C. (2010). On Over-fitting in Model Selection and Subsequent Selection Bias in Performance Evaluation. *Journal of Machine Learning Research*, 11, 2079–2107.
- Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., & Roth, A. (2015). Preserving Statistical Validity in Adaptive Data Analysis. *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC 2015)*.
- Ezharjan [GitHub handle; author name not stated in the repository] (2026). HallucinationPatternDetection: Hallucination Pattern Detection in Open-Source LLMs [Software]. GitHub repository, commit `ea0b967`. github.com/Ezharjan/HallucinationPatternDetection.
- Geifman, Y., & El-Yaniv, R. (2017). Selective Classification for Deep Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS 2017)*.
- Efron, B. (2011). Tweedie’s Formula and Selection Bias. *Journal of the American Statistical Association*, 106(496), 1602–1614.
- Fithian, W., Sun, D., & Taylor, J. (2014). Optimal Inference After Model Selection. *arXiv:1410.2597*.
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On Calibration of Modern Neural Networks. *International Conference on Machine Learning (ICML 2017)*.
- Gupta, C., & Ramdas, A. (2021). Distribution-Free Calibration Guarantees for Histogram Binning without Sample Splitting. *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*.
- Hussain, K., & Kantarcioglu, M. (2026). PARALLAX: Separating Genuine Hallucination Detection from Benchmark Construction Artifacts. *arXiv 2605.17028*.
- Kapoor, S., & Narayanan, A. (2023). Leakage and the Reproducibility Crisis in ML-based Science. *Patterns*, 4(9). *arXiv 2207.07048*.
- Kapoor, S., Cantrell, E. M., Peng, K., Pham, T. H., Bail, C. A., Gundersen, O. E., Hofman, J. M., Hullman, J., Lones, M. A., Malik, M. M., Nanayakkara, P., Poldrack, R. A., Raji, I. D., Roberts, M., Salganik, M. J., Serra-Garcia, M., Stewart, B. M., Vandewiele, G., & Narayanan, A. (2024). REFORMS: Consensus-based Recommendations for Machine-learning-based Science. *Science Advances*, 10(18). *arXiv 2308.07832*.
- Kumar, A., Liang, P. S., & Ma, T. (2019). Verified Uncertainty Calibration. *Advances in Neural Information Processing Systems (NeurIPS 2019)*.
- Li, Y., Fan, Z., & Zhuang, Z. (2026). When AUC 0.998 Is Not Enough: A Candidate Evaluation Protocol for Hidden-State Probes of Indirect Prompt Injection in Multimodal Computer-Use Agents. *arXiv 2606.22864*.
- Lee, J. D., Sun, D. L., Sun, Y., & Taylor, J. E. (2016). Exact Post-Selection Inference, with Application to the Lasso. *The Annals of Statistics*, 44(3), 907–927.
- Murphy, A. H. (1973). A New Vector Partition of the Probability Score. *Journal of Applied Meteorology*, 12(4), 595–600.
- Roelofs, R., Cain, N., Shlens, J., & Mozer, M. C. (2022). Mitigating Bias in Calibration Error Estimation. *Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS 2022)*.
- Roth, S. (2026). Which Leakage Types Matter? A Quantitative Landscape Across 2,047 Benchmark Datasets. *arXiv 2604.04199*.

Shi, W., Ajith, A., Xia, M., Huang, Y., Liu, D., Blevins, T., Chen, D., & Zettlemoyer, L. (2024). Detecting Pretraining Data from Large Language Models. *International Conference on Learning Representations (ICLR 2024)*.

Sun, L., & Bull, S. B. (2005). Reduction of Selection Bias in Genomewide Studies by Resampling. *Genetic Epidemiology*, 28(4), 352–367.

Vaicenavicius, J., Widmann, D., Andersson, C., Lindsten, F., Roll, J., & Schön, T. B. (2019). Evaluating Model Calibration in Classification. *Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics (AISTATS 2019)*.

Varma, S., & Simon, R. (2006). Bias in Error Estimation When Using Cross-Validation for Model Selection. *BMC Bioinformatics*, 7, 91.

Zadrozny, B., & Elkan, C. (2002). Transforming Classifier Scores into Accurate Multiclass Probability Estimates. *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD 2002)*.

Zhong, H., & Prentice, R. L. (2008). Bias-Reduced Estimators and Confidence Intervals for Odds Ratios in Genome-Wide Association Studies. *Biostatistics*, 9(4), 621–634.

Zöllner, S., & Pritchard, J. K. (2007). Overcoming the Winner’s Curse: Estimating Penetrance Parameters from Case-Control Data. *American Journal of Human Genetics*, 80(4), 605–615.

Guide to the Appendices

The appendices divide into four groups, and a reader checking a particular claim need only enter one.

The corrected procedure, applied and audited. Appendix M gives the three-line report of §4 for our own harness, the family-wise correction, and the permutation null we built and discarded.

What the numbers rest on. Appendix B states the resampling unit and what the intervals are about; Appendix D proves the two non-negativity results that make us decline the bootstrap elsewhere; Appendix Q defines every estimand used anywhere in the paper.

The audits themselves. Appendix C tabulates per-mechanism severity; Appendix F gives case-study detail; Appendix G quantifies the external ECE defect; Appendix H gives family verdicts; Appendix N holds the within-harness severity relationship.

Robustness, and where the paper fails its own checks. Appendix I holds the calibration robustness suite; Appendix J the reliability diagrams; Appendix E and Appendix K the regularization checks; Appendix L the second harness, in which the effect does not appear; Appendix P the prevalence study we did not run and why the obvious version would mislead; and Appendix R the extended limitations.

A Claim-to-Evidence Map

B Resampling Unit for All Bootstrap Intervals

Every BCa interval in this paper resamples the *replicate*, not the individual example. A replicate is one independent seed or fold-assignment draw; each contributes a single scalar (an AUROC gap, a Brier/ECE gap, or a selective-prediction risk), and BCa is applied to that vector with 10,000 resamples. Mechanism 1’s selective-prediction analysis uses 40 seeds; the remaining analyses use the replicate counts stated in Table 7. Two consequences follow. Reported risks and gaps are means over replicates, so a value such as Table 5’s 0.007 is a mean of per-replicate proportions and need not be attainable as a proportion at any single replicate’s sample size. And the intervals describe variability across seeds and fold assignments – the quantity these mechanisms actually perturb – rather than sampling variability across examples, which is the narrower question and would give tighter intervals than we report.

A. Documented – <i>descriptive; no effect size claimed</i>	
Five mechanisms	Present in the harnesses as written; three traced to pinned third-party source (§5).
B. Statistically established – <i>BCa 95% excludes zero; Holm-corrected where a family applies</i>	
M2 reuse gaps	+0.0255 AUROC, +0.0082 Brier, +0.0171 ECE between the reused fold and held-out data at the selected layer. Directly measured. <i>Not</i> attributed to selection: see Tier C.
M2 calibration (reuse)	Reliability +0.0046; ECE +0.0171 in all 8 binning settings; slope 0.68 → 0.62. A reuse gap, not a selection effect (§6).
Identified component <i>B</i>	+0.0173 AUROC ($t = 9.07$) and -0.0066 Brier (BCa $[-0.0087, -0.0041]$): the selected layer is genuinely better on data that played no part in selecting it (Appendix M).
M2 operating point M1, M3, M5	Risk inflated 10–16% of target at 50/70% coverage; 1,082 excess errors/day (§6.1). +0.13–0.29 (M1 reconstruction, 100 seeds); +0.0060–0.0067 (M3 real features); M5 ECE contamination reproduced externally (§O).
C. Illustrative, unresolved, or negative – <i>reported, not relied on</i>	
M1 narrative +0.1906	Prior work’s number; not reproducible from the artifact, excluded from every comparison.
M3 primary synthetic	Carried to -0.0001 by its own factorial control. The decomposition is the contribution, not the null.
M4	Non-negative by construction (Appendix D); n.s. vs. a permutation null, so its sign carries no information.
Selection attribution	Withdrawn. A placebo at unselected layers absorbs roughly half the gap, and the remaining selection-specific increment is tested against the wrong null; under the corrected test (§4) nothing is established on reliability or ECE. We claim no selection-specific calibration effect.
Cross-harness	Negative. On real features from two model families, selection inflates AUROC (+0.007, established in both) but the selection-specific calibration increment is null or negative (Appendix L). The calibration consequence does not follow from the discrimination one.
Scope limits	Calibration established on one feature representation (the four extra architectures test AUROC only); M2 at 30% coverage established only under the coverage-aware count; severity relationship diagnostic, not predictive (9.2–24.3× off); threshold transfer is informal, not conformal; per-day error counts scale one dataset to a hypothetical query volume rather than estimating a population; M5’s +0.0090 is quantified in our synthetic harness, not on the external pipeline’s own reported outputs; no prevalence estimate (repositories found by targeted search, not sampled).

Table 6: Claim-to-evidence map. A is what the code does; B is what the statistics support; C is what we report without relying on.

Sensitivity of the factorial attribution. The 75.5% selection-set-size share reported in §5 is computed at the shipped early-stopping fraction $F_{\text{small}}=0.15$. Re-running the factorial across $F_{\text{small}} \in \{0.05, 0.10, 0.15, 0.20\}$ (`results/es_hold_fraction_sensitivity_cs3_factorial.json`) gives shares of 74.7%, 69.0%, 75.5% and 38.5% respectively, against selection-run-budget shares of 16.6%, 31.4%, 28.1% and 62.5%. At the largest fraction the ranking inverts and the budget factor dominates. We therefore report selection-set size as the dominant factor at the shipped setting, not as a setting-independent decomposition; the uncorrected gap itself also shrinks monotonically across the sweep (0.0104 to 0.0024).

C Per-Mechanism Severity Table

The five mechanisms’ severity, each traced to source code. This is the paper’s self-audit of its own effect sizes, not its headline result: italicised qualifiers mark a number – ours or a source’s – that does not survive the control we built for it.

Mech.	Object selected / mechanism	Severity (code-verified)
1	Full-dataset probe fit, reused as feature – controlled synthetic reconstruction of the general mechanism	+0.13 to +0.29 AUROC, BCa-established, $n=100$ seeds
1	— the prior-work number (the degenerate case of selection: a candidate set of size one – the entire dataset – evaluated with no held-out complement)	+0.1906 AUROC – <i>not reproducible from this artifact</i>
2	CV-argmax layer selection	$\Delta_{\text{sel}} = +0.0255$ (SD 0.0186), randomized stratified splits; <i>single test, no within-family correction applies</i>
3	Per-fold checkpoint selection, fold-matched control	$\Delta_{\text{real}} = +0.0060$ to $+0.0067$ (real features); $+0.0159/+0.0044$ (synthetic, capacity 128/384)
3	— isotropic synthetic sweep, as-published vs. fully corrected control	$\Delta_{\text{synth}} = +0.0011$ to $+0.0036$ / -0.0001 – <i>not detectable once corrected</i>
4	Test-set best-layer selection, bootstrap max-bias	$+0.0021$ as computed, $+0.0029$ to $+0.0040$ bias-corrected (24 cells); <i>non-negative by construction, n.s. vs. permutation null</i>
5	Test-set threshold selection, independent validation split	AUROC: exactly 0 (algebraic); F1: $+0.022$ to $+0.052$ at $n=140$, $\approx 6.3\times$ smaller at $n=2000$

Table 7: The five mechanisms’ severity, each traced to source code – including one number we could not reproduce from the artifact, marked as such. Non-negativity proofs in Appendix D; derivations and factorial decompositions in the supplementary material, mapped in Appendix F.

D Non-negativity of Both Mechanism-4 Estimators

§5 states that Mechanism 4’s estimate rests on a statistic that cannot be negative by construction. Both statements and proofs are given here, so the claim is checkable without leaving this PDF.

Let $A \in \mathbb{R}^{L \times R}$ collect the AUROC of L probed layers under R seeds, with entries $a_{r,l}$. Write $m(l) = \frac{1}{R} \sum_r a_{r,l}$ for the grand mean at layer l , and

$$f(r, l) := m(l) - a_{r,l}, \quad \text{so that} \quad \sum_r f(r, l) = 0 \quad \text{for every fixed } l,$$

an identity holding for any data. In rotation r the selection set is the $R - 1$ seeds other than r , so the selection criterion at layer l is $c_r(l) = \frac{\sum_{s \neq r} a_{s,l}}{R-1} = m(l) + \frac{f(r,l)}{R-1}$, the rotation selects $L_r = \arg \max_l c_r(l)$, and its winner’s-curse term is $\text{wc}_r = c_r(L_r) - a_{r,L_r} = f(r, L_r) \cdot \frac{R}{R-1}$. Averaging over rotations,

$$\Delta_{\text{wc}} = \frac{1}{R} \sum_r \text{wc}_r = \frac{1}{R-1} \sum_r f(r, L_r). \quad (3)$$

Theorem 1 (Non-negativity of the rotation estimator). $\Delta_{\text{wc}} \geq 0$ for every $A \in \mathbb{R}^{L \times R}$, $R \geq 2$.

Proof. Let $L^* \in \arg \max_l m(l)$. Since L_r maximizes c_r , we have $c_r(L_r) \geq c_r(L^*)$, i.e. $m(L_r) + \frac{f(r, L_r)}{R-1} \geq m(L^*) + \frac{f(r, L^*)}{R-1}$, which rearranges to $f(r, L_r) \geq (R-1)(m(L^*) - m(L_r)) + f(r, L^*)$. Summing over r and using $\sum_r f(r, L^*) = 0$, $\sum_r f(r, L_r) \geq (R-1) \sum_r (m(L^*) - m(L_r)) \geq 0$, each summand on the right being non-negative because L^* maximizes m . By (3), $\Delta_{\text{wc}} \geq 0$. $\square$

The equal-argmax degeneracy is the equality case: if $L_r = L$ for all r then (3) gives $\Delta_{\text{wc}} = 0$ exactly, and in those 7 of 24 cells the estimator is not measuring a small effect, it is measuring nothing. `code/59` verifies the theorem numerically over 300,000 random matrices from five families, sweeping $L \in [2, 40]$ and $R \in [2, 8]$; the most negative value observed is -2.3×10^{-10} , float summation noise around an algebraic zero.

The replacement estimator is non-negative as well, by Jensen’s inequality rather than by cancellation. Write $\bar{m}(l) = \frac{1}{R} \sum_r a_{r,l}$, let $c^*(l)$ be the mean at layer l over a uniform bootstrap resample of the R seed indices,

and let $L^* = \arg \max_l c^*(l)$. The estimator `code/45` implements is

$$\Delta_{\text{boot}} := \mathbb{E}^*[c^*(L^*) - \bar{m}(L^*)] = \mathbb{E}^*\left[\max_l c^*(l)\right] - \mathbb{E}^*[\bar{m}(L^*)]. \quad (4)$$

Theorem 2 (Non-negativity of the bootstrap max-bias estimator). $\Delta_{\text{boot}} \geq 0$ for every $A \in \mathbb{R}^{L \times R}$, $R \geq 1$, with equality if and only if almost every resample’s argmax is also a grand-mean argmax and conversely — generically, if and only if the grand-mean argmax is bootstrap-stable.

Proof. A uniform bootstrap resample of the seed indices is mean-preserving, so for each fixed l , $\mathbb{E}^*[c^*(l)] = \bar{m}(l)$. The maximum is a convex functional of $(c^*(l))_l$, so by Jensen’s inequality $\mathbb{E}^*[\max_l c^*(l)] \geq \max_l \mathbb{E}^*[c^*(l)] = \max_l \bar{m}(l)$. Separately, $\bar{m}(L^*) \leq \max_l \bar{m}(l)$ for every realized L^* , so $\mathbb{E}^*[\bar{m}(L^*)] \leq \max_l \bar{m}(l)$. Substituting both into (4) gives $\Delta_{\text{boot}} \geq \max_l \bar{m}(l) - \max_l \bar{m}(l) = 0$. Equality forces both bounds to be tight simultaneously, which is the stated condition. $\square$

`code/70` verifies this over 50,000 random matrices at $R = 3$ and 2,500 at $R = 5$, by exact enumeration rather than sampling; the most negative value observed is -1.6×10^{-14} , again float residue. The two cells an earlier revision reported as negative are the equality case of Theorem 2: enumerating all 27 resamples at $R = 3$ gives -1.1×10^{-16} and $+3.3 \times 10^{-17}$ rather than -0.0001 and -0.0007 . The superseded values are retained in the shipped JSON as `bootstrap_max_bias_monte_carlo_legacy`.

E Mechanism 1’s Regularization Check

Mechanism 1’s probe is fit at $d = 414$, $n = 400$ and reaches exact in-sample separation at $C = 1.0$ (AUROC 1.0000), the overparameterized-interpolation regime, in which *any* classifier scored on its own training data looks reliable whether or not a selection step occurred. Repeating the calibration-bridge comparison at $C = 0.01$ (in-sample AUROC 0.997, no longer interpolating), the Brier gap survives but shrinks roughly 4×: +0.0061, BCa [+0.0058, +0.0063], against +0.0260 at $C = 1.0$. **The ECE gap does not survive:** +0.0002, BCa [−0.0004, +0.0010], against +0.0193. ECE at $n \approx 400$ with 10 bins is a bin-sensitive estimator (Kumar et al., 2019), and we read Mechanism 1’s ECE gap as an artifact of the near-interpolation regime it was measured in. Brier survives. Mechanism 2’s probe never interpolates, so none of its cells are affected.

F Extended Case Study Detail

Full per-case-study derivations, the complete $2 \times 2 \times 2$ factorial table, all 24 published cells behind Mechanism 4’s calibration, and a per-number provenance log recording each reported value and any earlier value it supersedes, are provided in the supplementary material (`EXTENDED_TECHNICAL_DETAIL.md`, `PROVENANCE_LOG.md`), together with a script-to-claim reproduction map (`REPRODUCE.md`). The full artifact is browsable at <https://anonymous.4open.science/r/selection-leakage-calibration-audit-XXXX> (anonymized mirror; the de-anonymized repository is linked on acceptance).

What the one-command verifier does and does not cover. `code/53_verify_paper_numbers.py` re-checks the severity, mechanism and severity-relationship numbers against their shipped result JSONs (558 checks, 0 failures at submission), and additionally asserts that every retracted value is absent from every shipped text artifact. It does *not* cover the calibration-bridge or selective-prediction tables (Tables 4 and 5); those numbers are produced by 95–97 and are checkable directly against their shipped JSONs, but not by that script. We state the boundary rather than let "code-verified" imply more coverage than it has; `REPRODUCE.md` gives the per-claim script-to-result map and a copy-pasteable rerun sequence. All of this is in the attached supplementary material, and the only GPU-dependent step is shipped as a cached array, so every check runs on CPU.

Family	Nom. sig.	Holm-surv.
Primary synthetic sweep, as-published control (4 cells)	2	1
Primary synthetic sweep, factorial-corrected control (4 cells)	0	0
Real-feature + fidelity extension, corrected control (4 cells)	2 (marg.)	0
Anisotropic sweep pooled w/ 7 other synthetic tests (8 cells)	≥ 2	2

Table 8: Every family from the severity analysis of §5–5 with a Holm-Bonferroni verdict, alongside its pre-correction picture; the calibration-bridge/selective-prediction family (Table 5) and the Mechanism-5 ECE family (§O) are reported separately. The final row’s nominal count is written as a lower bound because per-cell nominal significance was not recorded for all eight pooled cells; two is what the Holm-surviving set implies, not an exact tally. Two mechanisms rest on a single test or a permutation null rather than a multi-cell family, so no within-family correction applies to them.

G Quantifying the ECE Contamination (§O)

The pooled $+0.0090$ ECE gap reported in §O is quantified with the same already-validated Mechanism-5 synthetic harness used elsewhere in this paper (`code/46`, following the script-numbering convention in the supplementary material). Two details matter for interpreting it correctly.

First, the *honest baseline* in that harness is a threshold chosen on an independent validation split – not Table 4’s fold-based CLEAN_MATCHED refit. The two baselines answer different questions and should not be pooled.

Second, the gap is reported in Table 4’s subtraction order (honest minus leaky), so a positive value means the leaky condition looks *more* reliable than it honestly is – the same sign convention used throughout the paper.

The second audited Mechanism-5 repository (HallucinationPatternDetection, pinned commit `ea0b967`) does not report ECE at all, which is why this finding is single-source. We do not extend §6.1’s deployment-consequence analysis to it.

H Holm-Bonferroni Family Verdicts

Table 8 reports, for every family from the severity analysis of §5–5, how many cells were nominally significant and how many survive Holm-Bonferroni within that family. The calibration-bridge and selective-prediction cells (Tables 4 and 5) form a separate ten-test family, corrected jointly, of which nine survive.

I Calibration Robustness, Placebo Control, and the Coverage-Aware Error Count

Three analyses added after review, all recomputed from the cached per-sample probe scores of §6 (all 32 layers $\times$ 50 randomized splits are stored, so no new inference was required). Seed 20260925; BCa 95% intervals over the 50 splits as elsewhere. Script: `103_reviewer_response_mechanism2.py`, `104_deployment_error_count_corrected.py`. Appendix J gives the corresponding reliability diagrams and locates the miscalibration within the probability range.

(a) Brier decomposition, slope, and a binning-free calibration error. Brier score is a proper score, not a calibration metric: under the Murphy decomposition $BS = REL - RES + UNC$, only reliability is the calibration term, and an argmax over AUROC optimises resolution directly. Table 9 separates them. The effect sits in reliability, which is established; the resolution difference is not. Base rates are matched (0.7650

vs. 0.7633), so the uncertainty term is not driving the comparison. Two binning-free calibration statistics agree: the calibration slope moves further from 1 on held-out data, and the isotonic calibration error rises.

Quantity	LEAKY	CLEAN	Gap (BCa 95%)
Brier score	0.1491	0.1573	+0.0082 [+0.0030, +0.0136]
Reliability (REL)	0.0081	0.0127	+0.0046 [+0.0027, +0.0065]
Resolution (RES)	0.0379	0.0351	+0.0028 [−0.0014, +0.0070]
Uncertainty (UNC)	0.1798	0.1807	+0.0009
Calibration slope	0.6807	0.6185	+0.0621 [+0.0145, +0.1033]
Calibration intercept	0.0686	0.1385	—
Smooth cal. error	0.0065	0.0099	+0.0034 [+0.0020, +0.0050]

Table 9: Murphy decomposition at the selected layer, plus two binning-free calibration statistics. Reliability and resolution gaps are signed so that positive means “worse on held-out data”. Only the resolution interval covers zero. Slope 1 / intercept 0 is perfect; slope below 1 is overconfident. The isotonic fit is in-sample and so understates absolute calibration error equally in both arms; it is used only for the contrast.

(b) The ECE gap is not a binning artifact. Table 10 sweeps equal-width and equal-mass binning across {5, 10, 15, 20} bins. All eight settings exclude zero; the gap ranges +0.0164 to +0.0212, and the submitted +0.0171 sits mid-range rather than at a favorable extreme.

Binning	Bins	ECE LEAKY	ECE CLEAN	Gap (BCa 95%)
Equal-width	5	0.0623	0.0797	+0.0174 [+0.0102, +0.0247]
Equal-width	10	0.0716	0.0888	+0.0171 [+0.0099, +0.0241]
Equal-width	15	0.0774	0.0962	+0.0188 [+0.0127, +0.0256]
Equal-width	20	0.0833	0.1027	+0.0194 [+0.0126, +0.0266]
Equal-mass	5	0.0596	0.0760	+0.0164 [+0.0085, +0.0242]
Equal-mass	10	0.0666	0.0846	+0.0180 [+0.0111, +0.0257]
Equal-mass	15	0.0737	0.0922	+0.0185 [+0.0118, +0.0265]
Equal-mass	20	0.0799	0.1011	+0.0212 [+0.0145, +0.0289]

Table 10: ECE gap across eight binning settings. Every interval excludes zero. The equal-width 10-bin row is the one reported in the main text.

(c) Placebo control: how much of the gap is *selection*? The concern is that reusing a fold degrades calibration whether or not a layer was chosen on it, in which case attributing the gap to selection would overstate the mechanism. We therefore repeat the identical LEAKY-vs-CLEAN comparison at layers *not* chosen by the argmax – once averaged over all 32 layers per split, and once at a single uniformly drawn layer per split – and take the paired difference as the selection-specific increment.

Table 11 reports the outcome, and it is partly adverse to the straightforward reading. Both placebos are non-zero for the calibration metrics: fold reuse alone degrades reliability (+0.0020, interval excludes zero). The selection-specific increment remains established for every calibration metric, but it is roughly *half* the raw gap (56.6% of reliability, 57.9% of ECE, 43.6% of smooth calibration error), not all of it. We accordingly qualify the calibration claim in §6 and the abstract: selection contributes an established increment on top of an ordinary-reuse baseline that is itself real. Brier is the exception (91.8%), and that is consistent with (a): Brier mixes in resolution, which argmax-AUROC selection targets directly and which moves in the *opposite* direction at unselected layers – the resolution increment is significantly negative. This is a further reason not to read Brier as the calibration result.

Per-architecture selection-specific components (§6). The four additional architectures, each run with identical n , split, folds, reps, seed and probe, and with the machinery imported from the same script. K is the candidate count. These caches are forced-choice rather than generation-labeled, so their probe

Metric	Gap @ sel.	Placebo	Selection-specific (BCa 95%)	Share
Reliability	+0.00457	+0.00199	+0.00259 [+0.00135, +0.00397]	56.6%
ECE (10, width)	+0.01713	+0.00721	+0.00992 [+0.00541, +0.01496]	57.9%
Smooth cal. error	+0.00338	+0.00191	+0.00148 [+0.00044, +0.00265]	43.6%
Brier	+0.00819	+0.00067	+0.00752 [+0.00537, +0.00962]	91.8%
Resolution	-0.00280	+0.00201	-0.00482 [-0.00658, -0.00306]	—

Table 11: Placebo control. “Placebo” is the same LEAKY-vs-CLEAN gap averaged over all 32 layers rather than at the argmax; “selection-specific” is the paired difference. The single-random-layer placebo (not shown) gives the same qualitative picture with wider intervals and no placebo interval excluding zero. Resolution has no meaningful share because its gap at the selected layer is negative.

accuracies are not comparable with Mistral-7B’s; the selection-induced optimism measured on them is. They test the AUROC selection component only, not calibration.

Architecture	K	Δ_{sel}	p
GPT-2 medium	25	+0.0040	1×10^{-12}
Qwen-7B	29	+0.0008	0.076 (<i>n.s.</i>)
Qwen2.5-3B	37	+0.0048	3×10^{-17}
Qwen-14B	49	+0.0040	4×10^{-15}
Mistral-7B (ours, generation-labeled)	32	+0.0255	—

Table 12: Selection-specific AUROC component across architectures. Positive on all four; established on three at $\alpha = 0.05$. The sign carries; magnitude does not, and task and architecture change together across this comparison.

(d) Coverage-aware deployment error count. §6.1 originally multiplied the risk difference by a fixed answered-query count, which assumes the threshold transfers coverage as well as risk. It does not. Table 13 prices each arm at the coverage it achieves. The correction moves the headline figure from 972 to 1,082 errors per day and, at the 30% target where the coverage shift is largest, moves the interval from one covering zero to one excluding it. The correction is unfavorable to nothing in the paper; we retain the risk-difference verdict as the conservative statement and flag the 30% cell as established only under this estimator.

Target	Cov. LEAKY	Cov. CLEAN	Errors believed	Errors actual	Excess, corrected	Excess, submitted
30%	0.3000	0.3145	2,325	2,693	368 [77, 662]	249 [−39, 525]
50%	0.5000	0.5067	4,775	5,600	825 [412, 1260]	749 [339, 1173]
70%	0.7000	0.7071	8,945	10,027	1,082 [435, 1718]	972 [348, 1591]

Table 13: Expected errors per day at $Q = 100,000$ queries. “Believed” is $Q \cdot \text{cov}_{\text{LEAKY}} \cdot \text{risk}_{\text{LEAKY}}$; “actual” is $Q \cdot \text{cov}_{\text{CLEAN}} \cdot \text{risk}_{\text{CLEAN}}$ at the identical transferred threshold. “Submitted” is the fixed-coverage estimator $Q \cdot \text{target} \cdot (\text{risk}_{\text{CLEAN}} - \text{risk}_{\text{LEAKY}})$.

J Reliability Diagrams, and Where the Miscalibration Sits

The statistics in Appendix I say *that* the held-out predictions are less reliable; a reliability diagram says *where* in the probability range. Figure 2 pools all 50 splits’ predictions at each split’s own CV-argmax layer (20,000 reused-fold and 15,000 held-out predictions) into equal-mass bins, so a bin’s visual weight matches its statistical weight. Pooling mixes splits and is descriptive only: every inferential statement in this paper is computed per split and aggregated with BCa across splits.

Two features matter. Both arms are *under*-confident at the low end and *over*-confident above about $p = 0.6$, which is the familiar signature of a probe fit on few samples in high dimension. And the arms separate

in the high-confidence region: averaged over bins with mean predicted probability ≥ 0.8 – 10,500 held-out predictions, the region an abstention threshold actually operates in – the reused fold sits 0.0592 below the diagonal and the held-out data 0.0729, an excess overconfidence of 0.0136. That is the mechanical link between §6 and §6.1: the optimism is concentrated exactly where a selective-prediction threshold reads, which is why a gap of a few thousandths in aggregate ECE becomes a 10–16% inflation of the realized error budget.

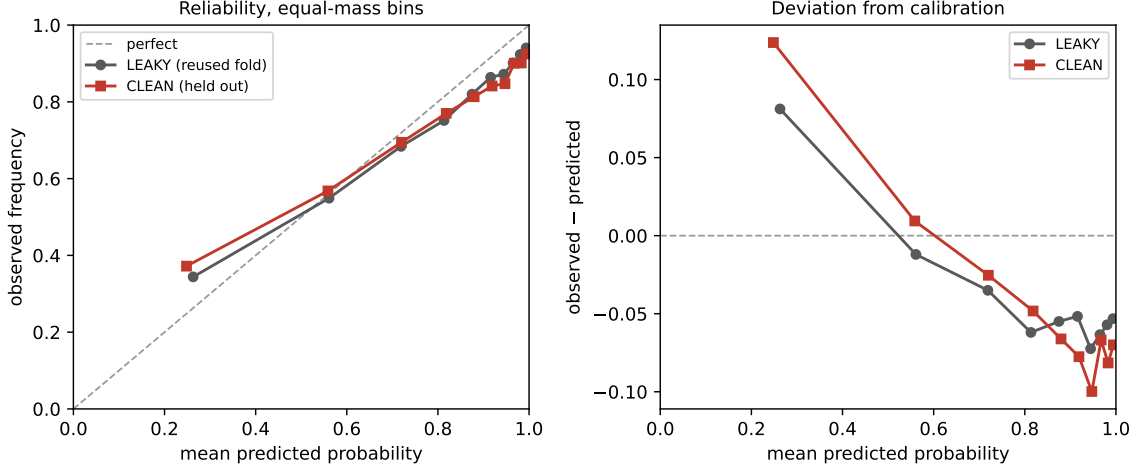


Figure 2: Reliability of the selected layer’s predictions, pooled over 50 splits, equal-mass bins. Left: observed frequency against mean predicted probability. Right: the same data as deviation from perfect calibration, where the separation is legible. Below the diagonal is overconfident. The held-out arm (red) is further below throughout the high-confidence region.

K Is the Calibration Effect an Interpolation Artifact?

Mechanism 1’s ECE gap does not survive stronger regularization (Appendix E), which raises the question of whether Mechanism 2’s calibration effect is likewise a consequence of fitting a $d=4096$ probe to $n=400$ samples near interpolation rather than of selection. The paper already answers this for *discrimination*: `code/48`’s sweep finds the selection-specific AUROC component stable across five decades of C . It had not been answered for calibration, which is the claim that matters here.

We therefore re-run the placebo decomposition of Appendix I under regularization: same 50 splits, same RNG stream, same 5-fold CV, varying only feature scaling and C . Table 14 reports it. **The selection-specific increment does not shrink under regularization; it grows.** At $C=0.01$ with standardized features – a regime that cannot interpolate – the reliability increment is $+0.0035$ (BCa $[+0.0024, +0.0048]$) against $+0.0026$ at the shipped probe, the ECE increment is $+0.0136$ against $+0.0099$, and the risk violation is unchanged at $+0.016$. What regularization removes is part of the *placebo*: the ordinary reuse baseline falls, so the share of the gap attributable to selection *rises* from 56.6% to 62.5%.

The shipped arm here is replayed from the cached artifact rather than refitted, so its row reproduces Appendix I exactly and serves as a consistency check on this script. Regularization is therefore not a competing explanation for Mechanism 2’s calibration result, and the contrast with Mechanism 1 is informative rather than awkward: Mechanism 1 fits on the evaluation data and its gap is an interpolation phenomenon, whereas Mechanism 2 never fits on the reported data at all and its gap is not.

L A Second Model Family: Where the Calibration Effect Does Not Replicate

The four extra architectures of §6 cannot address calibration: the artifact retains their summary statistics, not their per-sample scores. The real-feature harness can. It ships per-sample features for *two* model families

Probe	Metric	Gap @ sel.	Placebo	Selection-specific (BCa 95%)	Share
shipped $d=4096$, unscaled $C=1.0$	reliability	+0.00457	+0.00199	+0.00259 [+0.00135, +0.00397]	56.6%
	ECE	+0.01713	+0.00721	+0.00992 [+0.00541, +0.01496]	57.9%
	risk @50%	+0.01497	[+0.00679, +0.02347]		
scaled $C=1.0$	reliability	+0.00745	+0.00326	+0.00418 [+0.00227, +0.00633]	56.2%
	ECE	+0.01812	+0.00649	+0.01162 [+0.00709, +0.01646]	64.2%
	risk @50%	+0.01675	[+0.00872, +0.02480]		
scaled $C=0.01$ (no interpolation)	reliability	+0.00560	+0.00210	+0.00350 [+0.00239, +0.00480]	62.5%
	ECE	+0.02064	+0.00705	+0.01359 [+0.01015, +0.01739]	65.8%
	risk @50%	+0.01645	[+0.00733, +0.02567]		

Table 14: Selection-specific calibration effect under regularization, 50 splits each. Every selection-specific interval excludes zero in every probe configuration. `106_calibration_under_regularization.py`.

on the same pool – Mistral-7B and Qwen2.5-7B, HaluEval, the same extractor, 12 summary statistics at each of 32 sampled layers – so the identical Mechanism-2 construction can be run on both under a matched protocol. The probe there is 12-dimensional at $n=229$ and so cannot interpolate, which makes this a second, independent check on Appendix K’s question as well.

The result is **adverse to a general reading of our calibration claim**, and Table 15 reports it in full. In both model families the selection mechanism is plainly operating: Δ_{sel} on AUROC is positive and established (+0.0070 and +0.0072). But the selection-specific *calibration* increment is not positive in either. It is significantly *negative* for ECE in both (−0.0039, −0.0030) and for reliability in Mistral-7B (−0.0018), and the risk violation is not established in either.

We read this as a scope result, not a refutation. Magnitudes here are not comparable with §6’s – different features, different dataset, different n – so the comparison that carries information is one of sign and significance, and the sign does not carry. The honest statement is therefore narrower than “selection corrupts calibration”: **in the one additional harness we can test, selection inflates discrimination without corrupting calibration**. The calibration consequence is not an automatic corollary of the discrimination one, and a practitioner cannot infer it from the presence of a selection step alone. That is consistent with this paper’s own thesis about severity being a (mechanism, metric, control) triple; it adds that the harness belongs in the tuple too. It is also consistent with the paper’s existing statement that neither calibration gap is BCa-established on the real-feature harness – this sharpens that null into a measured, placebo-decomposed negative across two model families.

Model	Metric	Gap @ sel.	Placebo	Selection-specific (BCa 95%)
Mistral-7B	Δ_{sel} (AUROC)	+0.00700	[+0.00231, +0.01164] established	
	reliability	+0.00106	+0.00283	−0.00177 [−0.00284, −0.00078]
	ECE	+0.00779	+0.01171	−0.00392 [−0.00653, −0.00148]
	Brier	−0.00003	−0.00142	+0.00139 [−0.00120, +0.00402]
	risk @50%	+0.00212	[−0.00136, +0.00558] covers zero	
Qwen2.5-7B	Δ_{sel} (AUROC)	+0.00718	[+0.00542, +0.00915] established	
	reliability	+0.00102	+0.00171	−0.00069 [−0.00168, +0.00033]
	ECE	+0.00527	+0.00828	−0.00301 [−0.00567, −0.00004]
	Brier	−0.00045	−0.00302	+0.00257 [+0.00127, +0.00396]
	risk @50%	+0.00254	[−0.00187, +0.00681] covers zero	

Table 15: The Mechanism-2 construction on the real-feature harness, two model families, 50 splits each, matched protocol. Discrimination optimism replicates in both; the selection-specific calibration increment does not, and is significantly negative for ECE in both. `108_calibration_second_model_family.py`.

M Applying the Corrected Procedure to the Audited Harness

§4 prescribes a three-line report. Table 16 is that report for the Mechanism-2 harness, so that the paper practises what it prescribes. Selection is on the CV AUROC criterion in every row; only the metric being reported changes.

Metric	gap at l^*	A	B	B BCa 95%	t	Transfer?
AUROC	+0.0232	+0.0429	+0.0173	[+0.0134, +0.0209]	9.07	transfers
Reliability	−0.0046	−0.0038	−0.0012	[−0.0028, +0.0007]	−1.33	not established
ECE	−0.0171	−0.0106	−0.0007	[−0.0080, +0.0073]	−0.18	not established
Brier	−0.0082	−0.0142	−0.0066	[−0.0087, −0.0041]	−5.69	transfers

Table 16: The corrected report, 50 replicates. Here gap is written cv − ho, so for the three losses a *negative* value is the optimism (the selection fold looks better than held-out data); the main text reports the same quantities as held-out minus reused, hence with the opposite sign. B is tested against zero by one-sample t ; for a loss, $B < 0$ favours transfer. `112_corrected_report.py`.

Read this way the harness says something simple. Selection recovers a layer that genuinely transfers on AUROC ($t = 9.07$, $p = 5 \times 10^{-12}$) and on Brier ($t = -5.69$, $p = 7 \times 10^{-7}$), and we cannot establish transfer either way on reliability or ECE. The optimism the practitioner pays is the gap column, which needs no correction at all. At no point do we need Δ_{sel} , and had we tested it against zero we would have declared a selection effect on every metric, including the two where nothing is established.

Family-wise correction. These four B tests share a selection rule and a set of replicates, so they form one family and we correct them as we correct every other family in this paper. Under Holm-Bonferroni at family-wise $\alpha = 0.05$, AUROC ($p = 4.6 \times 10^{-12}$ against threshold 0.0125) and Brier ($p = 7.1 \times 10^{-7}$ against 0.0167) survive; reliability ($p = 0.190$) and ECE ($p = 0.861$) do not. **No verdict in Table 16 changes under correction.**

Why reliability and ECE are undetermined, and Brier is not. This is power, not a substantive difference between the metrics. The three losses point the same way, and their B estimates differ mainly in dispersion: the ECE interval is roughly five times as wide as the reliability interval and twenty times as wide as Brier’s, which is what one expects of a binned estimator at $n \approx 300$ per arm relative to a mean of squared errors. With 50 replicates, Brier clears zero, reliability does not quite, and ECE is not close. We read this as the binned estimator lacking resolution at these sample sizes, consistent with the estimator-bias literature cited in §6, not as evidence that calibration transfers while reliability does not.

A permutation null is redundant here, and we say so. We first approached this with a permutation that breaks the correspondence between each candidate’s selection-fold and held-out values, and `110_calibration_increment_null.py` in the artifact still computes it over 20,000 draws. It is redundant. Under that permutation $\mathbb{E}_\pi[\text{ho}_{\pi(l^*)}] = \overline{\text{ho}}$, so the null mean is exactly A ; by the identity $\Delta_{\text{sel}} = A - B$, the event “ Δ_{sel} falls below its permutation null” is the event “ $B > 0$ ”. The simulation therefore recovers, with Monte Carlo error, a conclusion the t -test gives exactly, and it additionally assumes candidates are exchangeable, which adjacent transformer layers are not. We retain the script so the equivalence can be checked — its simulated null means match A to 7×10^{-5} — and base nothing on it.

N The Within-Harness Severity Relationship

Within one harness, severity rises monotonically and concavely with candidate count K at a fixed operating point, from +0.0093 at $\text{AUROC}_0 = 0.70$ to indistinguishable from zero at 0.985 ($p = 0.219$, $n = 100$). This K -regularity is scoped to checkpoint-along-a-trajectory candidates only: an independently-seeded, exchangeable-candidate replication at adequate power ($N = 200$) does not reproduce it (gaps +0.0013 to −0.0010, every CI covering zero). The operating-point relationship is not similarly scoped but does not transport across

harnesses – it is exceeded by roughly an order of magnitude at a matched operating point in our own real-feature harnesses, which is the $9.2\text{--}24.3\times$ figure quoted in §7.

Does mechanism identity matter beyond these two covariates? A matched comparison at a common (K, AUROC_0) shows a per-mechanism-intercept model beats a pooled one decisively ($304\times$ intercept spread), and a pooled fit without a mechanism term gives a wrong-signed $\ln K$ coefficient that flips once mechanism is controlled for. Some factor beyond candidate count and operating point is therefore not absorbed by them – though this design cannot separate whether it is mechanism identity itself or the unequal harness maturity correlated with it. What this licenses is only the negative conclusion: severity is not governed by those two covariates to the exclusion of mechanism. We do not read the severity surface as a calculator, and §7 says so.

O The Externally Verified ECE Defect

Unlike §6’s comparisons, which we construct on our own cached scores, this is a bug we *found* in MultiHaluDet’s code (a multilingual hallucination-detection pipeline; Alvi et al., 2026; pinned commit c759751). Its ECE term uses `confidence=max(probs,1-probs)` (threshold-independent) but `correct=(preds==labels)`, with `preds` thresholded at Mechanism 5’s own test-label-selected Youden cut: **the pipeline’s own reported ECE is contaminated by the mechanism this paper audits it for.** Quantified with the already-validated Mechanism-5 synthetic harness (Appendix G), the pooled ECE gap across five operating points is $+0.0090$, all five cells BCa-established. This is a code-level defect rather than a decomposed attribution, so the null issue of Appendix M does not bear on it; but it is verified in one of the two audited repositories only, and we do not generalize from one instance.

P What a Prevalence Study Would Require

Two reviews of this work asked how widespread these mechanisms are. We give no estimate, and the reason is worth stating precisely, because the cheap version of this study would produce a confidently wrong number.

There is no denominator. The repositories here were found by targeted adversarial search for this failure class. That yields no sampling frame, so the audited set supports neither a rate nor even a lower bound: a search designed to find instances tells you nothing about how often you would find them if you had not been looking.

The obvious instrument is disqualified by our own measurement. §7 reports what happened when we automated four of the five checklist items with a regex scanner: 7 raw hits across 7 repositories, *all* false positives, and *both* known true positives missed. Its measured precision and recall on this failure class are therefore 0 and 0. Running that scanner across a large corpus would return a number, and the number would be an undercount with no error bars. A prevalence study that scaled the cheapest available instrument would be exactly the kind of confidently-wrong evaluation this paper exists to audit.

What a valid design needs. Four things, in order. (i) A sampling frame: an enumerable population, such as every paper in a fixed set of venues and years that reports a hidden-state probe metric *and* releases code, since a mechanism that cannot be checked against source cannot be scored. (ii) A random sample from that frame, not a search. (iii) An instrument with measured error rates on this failure class. Our blinded language-model re-read is the only candidate we have evidence for – it agreed with all 7 manual classifications and located both missed bugs unprompted – but agreement on 7 cases is a pilot, not a validation. (iv) A human-adjudicated validation subsample large enough to estimate that instrument’s sensitivity and specificity with intervals, so the raw hit rate can be corrected for misclassification rather than reported as a prevalence.

Only step (iv) makes the resulting number interpretable, and none of the four is expensive relative to the cost of getting the answer wrong. We flag this as the single most valuable follow-up to this paper, and as the reason its taxonomy and checklist – not any severity number – are its most reusable output.

Q Estimands Used in This Paper

Reviewers noted that the number of distinct estimands, controls and qualifications makes the paper dense. Every quantity we report is one of the following six; nothing else is estimated.

- **Optimism gap** = metric on the data used for selection – metric on genuinely held-out data, at the same fixed configuration. The base quantity all others derive from.
- **Selection-specific component** = optimism gap at the layer the argmax *would select* – mean optimism gap across all candidate layers. Positive in expectation even under a pure-noise null, by construction: that is the winner’s curse it exists to measure, and it is why its significance must not be read as evidence of any further pathology (Appendix C).
- Δ_{sel} = the selection-specific component measured on AUROC specifically. The paper’s discrimination headline.
- **Calibration gap** = optimism gap measured on Brier, its Murphy reliability term, ECE, or isotonic calibration error. Signed so positive means worse on held-out data.
- **Placebo gap** = the same LEAKY/CLEAN comparison at layers the argmax did *not* select, and **selection-specific increment** = calibration gap at the selected layer – placebo gap (Appendix I).
- **Risk violation** = risk achieved on held-out data at a threshold transferred from the reused fold – risk that threshold reported on the reused fold. The operational quantity.

Two controls recur. **CLEAN_MATCHED** refits on the selection pool and scores genuinely held-out data, matching the leaky arm’s training size. **Fold-matched** controls equalise selection-set size between arms, which is the correction that carries Mechanism 3’s primary estimate to null.

R Extended Limitations

This appendix states in full the limitations summarised in the main text; no limitation is stated here that is not also flagged there.

Which mechanisms the calibration findings cover. The calibration-corruption and selective-prediction findings (§6–6.1) are checked on three of the five mechanisms in this paper’s taxonomy and established on two, because only those three had per-sample probability outputs already cached; we do not claim either finding holds for Mechanisms 4–5 without directly checking each, and Mechanisms 4 and 5’s calibration behavior remains unchecked here. §O’s ECE-contamination finding is a separate, code-verified bug, not a check of Mechanism 5’s general calibration behavior, and we do not extend §6.1’s deployment-consequence analysis to it. The third of the three, where the effect is not established, is Mechanism 3’s real-feature harness (its own MLP and checkpoint-selection pipeline, both capacities). Neither Brier nor ECE gap is BCa-established there (both capacities’ intervals include zero), consistent with that harness’s own AUROC effect being only marginal-to-null after correction, and not evidence against the pattern found elsewhere.

Interpolation regime and the regularization sweep. Mechanism 1’s probe operates near the interpolation threshold ($d = 414$, $n = 400$) at its default regularization. A regularization-sweep robustness check shows the risk-violation finding is stable to this (if anything marginally larger away from interpolation), the Brier-score gap survives but shrinks roughly 4×, and the ECE gap specifically does not survive at stronger regularization – we report all three outcomes rather than the single most favorable one.

Relationship to distribution-free coverage guarantees. The selective-prediction analysis is an informal illustration of the question Bates et al. (2021) treat formally, not a construction or violation of a specific conformal-prediction method’s distribution-free coverage guarantee. One of ten new statistical cells across both findings (Mechanism 2 at 30% target coverage) is not established at the per-cell stage; the family-wise correction changes no verdict among the other nine. Throughout, we report null and marginal results alongside positive ones rather than selectively reporting favorable cells.

Prevalence, transfer, and scope. We give no prevalence estimate for these mechanisms across the literature: repositories were found by targeted search for this failure class, not sampled, so the taxonomy’s coverage of the wider literature is unknown. The severity surface (§5) is an empirical fit, not a bound, and whether its estimates transfer to real hidden states beyond the checks already run is untested. The candidate-count regularity is scoped to checkpoint-along-a-trajectory selection and does not reproduce for exchangeable candidates at adequate power – readers should not extrapolate it to layer or threshold selection without re-measuring. One case study’s headline number is not independently re-verifiable from this artifact and is excluded from every quantitative comparison. The matched five-mechanism comparison (§5) is a controlled-harness finding across reconstructions of unequal maturity, not a claim about the five real audited pipelines at a matched operating point.

Pre-registered decision rule and its family-wise consequence. This paper’s pre-registered rule is the unadjusted per-cell BCa interval; under family-wise (Holm-Bonferroni) correction, essentially no marginal positive finding survives outside one mechanism’s non-marginal threshold-selection result and the exceptions in Table 8 (two families, three cells). We report this consequence explicitly because it changes how every marginal claim in §5 should be weighed.

The negative automation result. The false-positive rate is measured across all 7 repositories found by targeted search; the miss rate is measured only against the 2 with known ground-truth leaks. Neither is a random or exhaustive sample, and does not establish that this leakage class resists automated detection in general, only that line-level regex matching is the wrong instrument.

What we propose as a baseline. We position the disclosure practices used throughout this paper – naming a decision rule before reporting against it, reporting corrections to our own prior estimates, and stating which claims we cannot verify from this artifact – as a baseline other evaluation-leakage audits in this space can adopt, not as a substitute for the prevalence study this paper does not run.